Srivathsa Mantrala

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Education

North Carolina State University, Raleigh, NC Aug. 2023 – May 2027
***B.S.** Computer Science (Artificial Intelligence), **Minor** in Business Administration* ***GPA:** 3.55 / 4.0*
Relevant Coursework: Software Engineering, Data Structures, Artificial Intelligence, Automated Learning & Data Analytics, Linear Algebra, Probability & Statistics, Computer Security, Info Systems Management

Technical Skills

Languages: Java, Python, C, SQL, JavaScript, TypeScript, HTML/CSS
Frameworks & Testing: Spring Boot, React, Node.js, Flask, FastAPI, JUnit
Databases: PostgreSQL, MongoDB, ChromaDB
Libraries: React Flow, Axios, pandas, NumPy, Matplotlib, Scikit-learn
Developer Tools: Git, GitHub Actions, Docker, Maven, IntelliJ IDEA, VS Code, JupyterLab, OpenAI API

Experience

TA - CSC 201: Fundamentals of Applied AI Jan. 2026 – Present
North Carolina State University *Raleigh, NC*
- Lead 2 weekly lab sessions for 40+ students across 15+ engineering majors, teaching applied AI and machine learning concepts through hands-on Python and JupyterLab exercises.
- Debug student Python and ML code, answer technical questions, and provide feedback on assignments, helping students resolve data-handling and analysis issues over 14 in–lab tutorials.

Operations Manager - NC State Student Centers Aug. 2023 – Present
North Carolina State University *Raleigh, NC*
- Coordinate operations for a 250,000+ sq. ft. student facility, leading teams of 4–10 members per shift and supporting 20+ events per week, and training new staff to help run shifts independently.

Projects

VisualSQL | *Java, Spring Boot, React, TypeScript, JUnit, JSqlParser, Maven* Jul. 2026 – Present
- Designed a Spring Boot backend that parses SQL DDL into graph-based ER models using JSqlParser.
- Implemented a modular parsing pipeline for primary keys, foreign keys, relationships, and DTO mapping exposed through a REST API.
- Wrote 15 JUnit tests with GitHub Actions CI, achieving ∼95% coverage of the SQL parsing pipeline (73% overall).

JobTrail | *Java, Spring Boot, React, PostgreSQL, JWT, OpenAI* Mar. 2026 – Apr. 2026
- Developed a full-stack job application tracker with JWT authentication, PostgreSQL persistence, and resume management.
- Built REST APIs with 11 endpoints for application tracking, analytics, and AI-assisted job posting parsing using GPT-4o-mini.
- Integrated Cloudinary, PDFBox, and React dashboards to manage resume versions and visualize job search analytics.

TruthTrace | *Java, Spring Boot, OpenAI API* *Hack_NCState (Team of 3)* Feb. 2026
- Engineered a URL credibility analyzer that flags phishing, scams and manipulative content using 20 heuristic checks across domain structure, domain age, HTTP behavior and article text, producing a 0–100 threat score.
- Integrated WHOIS domain-age lookups to flag newly registered domains, and an article-extraction API to scan page text for urgency, credential-harvesting phrases and clickbait.
- Generated natural language explanations with GPT-4.1-mini, based on the pipeline's detected signals.

JGaffer | *Java, Python, Spring Boot, React, TypeScript, FastAPI, ChromaDB, PostgreSQL* Jan. 2026 – Present
- Designed a three-tier architecture with Spring Boot, React, and FastAPI to power AI-driven football tactics analysis.
- Engineered a RAG pipeline using ChromaDB and GPT-4o-mini to generate grounded tactical explanations from football match data.
- Built an interactive tactical puzzle engine using graph visualizations and passing success prediction model to evaluate passing options in real time.